

Multi-Probe Evidence for an Environment-Dependent Cosmological Transition at $z < 0.04$

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Abstract

Four distinct low-redshift probes converge on a common environment-dependent transition at $z = 0.040 \pm 0.005$ (comoving distance ~ 170 Mpc). This scale matches the measured radial extent of the local underdensity and is about 7.5 times the strain-lever length scale $\beta = 22.7$ Mpc that the Mesche Tensor Dynamics Framework (MTDF) constrains independently from void quantisation; the quantitative link between β and the transition scale is a conjecture of the framework rather than a direct equality. The convergence is summarised in the table below.

Table 1. Multi-probe convergence at the ~ 170 Mpc low-redshift transition scale.

Probe	Dataset	Transition scale	Key statistic
SNe piecewise	Pantheon+ merged (3 void finders)	$z = 0.040 \pm 0.005$	$\Delta\chi^2 = 36\text{--}39$, $p < 0.001$
Peculiar velocity	Cosmicflows-4 (VoidFinder, REVOLVER, VIDE)	Signal concentrated at $z < 0.04$	9.0σ N-body tension (MDPL2, 100 observers; nominal, error-model conditional)
Tully-Fisher residuals	2MTF (J, H, K bands)	Entire sample $z < 0.033$ (in-regime)	4.2σ N-body tension, achromatic
Time-delay distances	H0LiCOW strong lenses	Geometric, local H_0 ($S_0 = 1.084$)	Consistent with MTDF local H_0 (equivalent to the established local- H_0 preference; not independent evidence)

Multiple independent low-redshift observables are tested for a common environment-dependent transition near a single low-redshift transition scale (comoving ~ 170 Mpc). The Mesche Tensor Dynamics Framework (MTDF) predicts that such deviations should concentrate below $z \approx 0.04$, the measured extent of the local underdensity (about 7.5β ; the link to β is a structural conjecture). Four channels provide direct evidence; they are best read as probes of one environment-linked phenomenon, since the channels share the local large-scale structure field, and CF4 itself contains SNIa distances. Supernova luminosity distances show an environment signal ($p = 0.003$) with a sharp piecewise cutoff at $z = 0.04$. Cosmicflows-4 peculiar velocities reveal a void-geometric coupling that survives a seven-layer control chain and exhibits an amplitude inconsistent with the near-zero N-body expectation, with a sign reversal relative to linear theory (9.0σ tension, nominal, error-model conditional; MDPL2, 100 observers). 2MTF Tully-Fisher residuals independently confirm the pattern, achromatic across three infrared bands, with 4.2σ N-body tension. An additional geometric support channel, strong-lens time-delay distances, is consistent with the MTDF local H_0 (equivalent to the established local- H_0 preference, not independent evidence). Two further channels (ISW void stacking, weak-lensing S_8 splits) establish methodology and show

directional consistency but remain below detection threshold with current data. The convergence of independent probes at a single transition scale motivates targeted independent replication and dedicated Λ CDM systematics tests of the same environment-dependent observables.

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Claim status. This paper distinguishes between: (i) calibration anchors, (ii) validation targets, (iii) diagnostic consistency checks, and (iv) speculative extensions. Only results explicitly labelled as validation targets are used as evidence for the core MTDF claim. Diagnostic and speculative sections are included to define future tests, not to claim established confirmation.

Conventions. This paper is written against the frozen unified action (MTDF_01, Section 2A). Claim statuses follow the frozen eight-class ledger; growth-sector statements defer to the sealed v2 posture (enhanced growth; the former S_8 -relief reading is retired as a gauge artifact; see MTDF_01 Section 7.3). The strain-lever length scale $\beta = 22.7$ Mpc was formerly labelled a coherence length; the label is retired, the number and role are unchanged.

Nominal significance. Unless explicitly stated otherwise, quoted significances are conditional on the specified catalogue, control chain, redshift cut, and metric definition. They should be read as nominal or pipeline-conditioned significances, not as final global discovery significances after all possible analysis choices. A full look-elsewhere assessment across alternative void metrics, redshift partitions, catalogue choices, and environment definitions remains a priority for independent replication.

1 Introduction

Standard cosmological analyses typically treat low-redshift probes independently. Supernova distance moduli, peculiar velocity fields, galaxy scaling relations, and CMB secondary anisotropies are each analysed within their own pipelines, subject to their own systematics, and compared against Λ CDM expectations in isolation. Environment is usually treated as a nuisance variable to be marginalised over rather than a carrier of physical information.

This paper takes a different approach. It tests whether several independent low-redshift observables point to a common environment-linked transition concentrated near a single redshift scale. If multiple probes, each with different physics, different systematics, and different survey selection functions, converge on the same transition regime, this constitutes a physical clue that cannot easily be attributed to any one dataset or method.

The motivation arises from the Mesche Tensor Dynamics Framework (MTDF), which specifies a characteristic transition redshift $z \approx 0.04$ (the quantitative link to its strain-lever length scale $\beta = 22.7$ Mpc is a structural conjecture). Below this scale, the MTDF stress tensor field couples to observables in an environment-dependent manner; above it, the effect averages out over many strain-lever lengths along the line of sight. The length scale β was fixed in MTDF_01, and the $z < 0.04$ localisation was identified in the supernova channel (December 2025) before the velocity-field tests reported here were conducted; Section 2 gives the dated record of this ordering.

The paper is structured as follows. Section 2 states the working hypothesis and its quantitative prediction. Section 3 describes the six datasets. Section 4 presents the shared analysis logic applied uniformly across all channels. Section 5 reports the results for each channel with its associated controls. Section 6 synthesises the cross-probe evidence. Section 7 presents the Λ CDM mock comparison, including full N-body results. Sections 8 and 9 discuss interpretation and limitations. Section 10 concludes.

2 Working hypothesis and prediction

MTDF introduces a stress tensor field with the finite strain-lever length scale $\beta = 22.7$ Mpc. The environment-linked transition is observed at a larger scale, a comoving distance of approximately 170 Mpc ($z \approx 0.04$), which is about 7.5 times β ; the quantitative relation between the strain-lever length scale and this transition scale is a conjecture of the framework rather than a direct equality, with the transition scale matching the measured radial extent of the local underdensity. The framework predicts that environment-dependent deviations from Λ CDM should exhibit a scale hierarchy governed by the dimensionless product βk , where k is the characteristic wavenumber of the observation:

$$\beta k \ll 1 \Rightarrow \text{effects active (cosmological scales)}, \quad \beta k \gg 1 \Rightarrow \text{effects suppressed (local scales)}$$

The specific, falsifiable prediction tested in this paper is:

Prediction. Environment-dependent residuals in distance indicators and velocity fields should be (i) present below $z \approx 0.04$, (ii) absent or strongly suppressed above $z \approx 0.04$, and (iii) correlated with the signed distance to the nearest void boundary as classified by independent void-finding algorithms applied to galaxy redshift surveys.

The dated record for this ordering is as follows. The length scale itself has been fixed in the MTDF parameter set since the earliest archived versions of MTDF_01 ($\beta = 22.685 \approx 22.7$ Mpc, project archive of December 2025 to February 2026), together with the scale factor $S_0 = 1.084$. The $z < 0.04$ localisation was first identified in the supernova channel: the pre-registered SN $\times$ void analysis script and its provenance manifest are dated 2025-12-16 and 2025-12-17 respectively, several months before the CF4 and 2MTF velocity-field analyses of Sections 5–7 were run (April 2026). No dated artefact establishes $z \approx 0.04$ as a theoretical pre-specification prior to that December 2025 supernova analysis; the transition scale therefore functions as a genuine advance prediction for the velocity-field channels, and as a data-identified localisation within the supernova channel. The prediction for the velocity-field channels is sharp: a smooth or absent transition would falsify the finite-range interpretation of the transition scale.

3 Datasets and observables

Six independent observables are tested. They span different physical processes, survey instruments, galaxy populations, and redshift ranges within the low- z universe, and the channels are largely independent in calibration and host-galaxy properties; the one known overlap is that the CF4 compilation contains 945 SNIa distances (see Section 6). They are, however, sensitive to the same underlying low-redshift velocity and distance field; their convergence is therefore best read as multiple independent measurements of one environment-linked phenomenon, not as several separate phenomena. This strengthens the case that the phenomenon is real while sharpening the requirement to identify its mechanism (MTDF stress depletion versus an unmodelled velocity-field systematic common to all distance indicators).

3.1 Supernova luminosity distances

The merged sample comprises 4,188 Type Ia supernovae from Pantheon+ (1,533 SNe with full STAT+SYS covariance), ZTF DR2 (2,650 SNe, Tripp-standardised), and Foundation DR1 (5 unique SNe after deduplication). Distance modulus residuals are computed after SALT2 standardisation. Environment classification uses DESI DESIVAST void catalogues cross-matched at 5 arcsecond tolerance. The low-redshift subsample at $z < 0.04$ provides the primary test regime. This channel was established in MTDF_02 and the GPU validation paper MTDF_07; it is summarised here for completeness.

3.2 Strong-lens time-delay distances

Seven multiply-imaged quasar systems from the TDCOSMO/H0LiCOW programme provide time-delay distance measurements that are geometrically independent of supernova luminosity distances. An additional 25 SLACS lens systems provide velocity dispersion-based distance estimates at lower redshifts ($z_L = 0.08\text{--}0.35$), where DESIVAST void coverage is more complete.

3.3 Cosmicflows-4 peculiar velocities

The Cosmicflows-4 catalogue (Tully et al. 2023) contains 55,877 galaxies in 38,053 groups at $z < 0.05$, with distances measured via eight independent methods (primarily Tully-Fisher for spirals and Fundamental Plane for ellipticals). Group-averaged peculiar velocities are used to reduce individual measurement scatter. The catalogue is accessed from the Extragalactic Distance Database (EDD). Environment classification uses the same DESIVAST void catalogues as the supernova analysis.

3.4 2MTF Tully-Fisher distances

The 2MASS Tully-Fisher survey (Hong et al. 2019) provides Tully-Fisher distances for 2,062 nearby spiral galaxies at $600 < cz < 10,000$ km/s ($z < 0.033$). Distances are measured in three independent infrared bands (J, H, K) from 2MASS photometry, combined with HI 21 cm linewidths. The entire sample falls below $z = 0.04$, placing it entirely within the predicted MTFD-active regime.

3.5 ISW void stacking

The integrated Sachs-Wolfe (ISW) effect is tested via stacked CMB temperature profiles centred on voids from the BOSS and DESIVAST catalogues. Planck PR4 SMICA temperature maps provide the CMB signal. A total of 889 BOSS voids and 563 DESIVAST voids (1,452 combined) are used. MTFD predicts a specific void radial profile shape that differs from the step-function template commonly used in ISW analyses.

3.6 Weak-lensing S_8 environment split

Galaxy-galaxy lensing (GGL) tangential shear profiles are measured from KiDS-1000 source galaxies around GAMA spectroscopic group lenses, split by foreground DESIVAST void classification. The analysis processes 2.7 billion source-lens pairs across 1,724 void-embedded and 80 wall-embedded galaxy groups.

4 Common analysis logic

All channels are evaluated under a shared methodological framework. This consistency is essential: without it, multi-probe convergence could be an artefact of inconsistent analysis choices rather than a physical signal.

4.1 Signed-distance environment metric

Environment is quantified using a continuous signed-distance metric rather than a binary void/wall classification. For each object, the signed distance d_{signed} to the nearest DESIVAST void boundary is computed as:

$$d_{\text{signed}} = \frac{r_{\text{obj}} - R_{\text{void}}}{R_{\text{void}}}$$

where r_{obj} is the distance from the object to the void centre and R_{void} is the void radius. Negative values indicate positions inside a void; positive values indicate wall or filament environments. This

continuous metric retains more information than a binary split and enables gradient analysis even when one environment class has few or zero members.

4.2 Three independent void catalogues

All analyses are repeated with three independent void-finding algorithms applied to the DESI Bright Galaxy Survey: VoidFinder, REVOLVER, and VIDE. These algorithms use different void definitions, identification procedures, and boundary conventions. Consistency across all three provides robustness against void-finder-specific artefacts.

4.3 GLS regression framework

Each channel is tested using generalised least-squares (GLS) regression of the observable residual against d_{signed} :

$$y_i = \alpha + \gamma \cdot d_{\text{signed},i} + \varepsilon_i$$

where y_i is the residual for the i -th object (distance modulus residual, peculiar velocity residual, or TF residual), γ is the environment coupling coefficient, and the covariance structure is channel-specific. The null hypothesis is $\gamma = 0$.

4.4 Piecewise redshift split

For channels with sufficient redshift range, a piecewise model splits the sample at $z = 0.04$ and fits separate γ values above and below the cut. The $\Delta\chi^2$ between the piecewise model and the single- γ model quantifies the evidence for a redshift-dependent transition. This directly tests the transition-scale prediction.

4.5 Robustness chain

Each channel is subjected to a seven-layer control chain (layers applied where applicable):

Table 2. Table 1. Standard control chain applied to each observable channel.

Layer	Control test	Question addressed
A	Density substitution	Is d_{signed} simply a proxy for local galaxy number density?
B	Partial correlation	Does d_{signed} carry information beyond local density?
C	Λ CDM chain rule	Is the observed coupling consistent with the density-observable relation predicted by Λ CDM?
D	2M++ velocity-field subtraction	Does the signal survive after removing the best Λ CDM velocity reconstruction?
E	Malmquist bias test	Is the signal driven by distance-dependent selection effects or catalogue geometry?
F	Leave-one-method-out (LOSO)	Does the signal depend on a single distance-method family?
N-body	MDPL2 mock (100 observers)	Can a full non-linear Λ CDM simulation reproduce the signal?

Layers E and F are applied where the dataset combines multiple distance methods (CF4). Additionally, all channels are checked for footprint consistency (NGC vs SGC) and permutation significance (10,000 shuffles of d_{signed} labels with 5,000 bootstrap resamples).

5 Channel results

5.1 Supernovae: environment cutoff at $z = 0.04$

The supernova environment signal was established in MTDF_02 and independently validated in MTDF_07. On the merged 4,188-SN sample, a piecewise model with a break at $z = 0.04$ yields $\Delta\chi^2 = 36\text{--}39$ (diagonal-covariance, nominal; $p < 0.001$) compared to a single-regime fit. The signal is concentrated in the low-redshift bin:

Table 3. Table 2. Supernova environment signal by redshift regime (Pantheon+ $\times$ DESIVAST).

Regime	γ_{env}	Significance	Void catalogues
$z \in [0.02, 0.04)$	$+0.013 \pm 0.004$	$p = 0.003\text{--}0.005$	REVOLVER, VIDE
$z > 0.04$	Consistent with zero	$p > 0.84$	All three

Control outcome. The piecewise cutoff at $z = 0.04$ was carried over from the December 2025 supernova analysis (dated provenance manifest, 2025-12-17) and was therefore fixed before the velocity-field analyses of Sections 5–7; within the supernova channel itself, the cutoff was identified by the redshift-binned analysis rather than specified ahead of it. The signal passes LOSO (16/16 surveys retain positive γ_{env}), permutation tests ($p_{\text{perm}} = 0.025\text{--}0.062$), block bootstrap (95% CI excludes zero for REVOLVER and VIDE), and survey fixed effects (γ_{env} stable within 0.0005 mag). SALT2 population controls shift γ_{env} by less than 0.3σ .

5.2 Time-delay distances: independent geometric support

This channel differs in kind from the other probes: it does not test the void-wall environment split directly but provides an independent geometric consistency check on the MTDF low-redshift interpretation. Seven TDCOSMO/H0LiCOW lenses provide time-delay distance measurements that test the MTDF scale factor $S_0 = 1.084$, which maps the Planck-calibrated $H_0 = 67.36$ km/s/Mpc to an effective local value of $H_0 \approx 73.1$ km/s/Mpc.

Table 4. Table 3. Time-delay distance goodness of fit: MTDF vs Λ CDM.

Model	χ^2/dof	Mean residual	Residual pattern
ΛCDM ($H_0 = 67.36$)	$26.05/7 = 3.72$	-1.58σ	All 7 negative (systematic)
MTDF ($H_0 = 73.1$ via S_0)	$4.77/7 = 0.68$	-0.10σ	Mixed $+/-$ (unbiased)

The MTDF χ^2/dof is a factor of 5.5 lower than the Planck ΛCDM value; this is consistent with the MTDF local H_0 and is equivalent to the established local- H_0 preference, not independent evidence. The ΛCDM residuals are systematically negative across all seven lenses, indicating a coherent underprediction of time-delay distances. The MTDF residuals scatter around zero with no systematic bias.

An environment correlation test on the TDCOSMO lenses is inconclusive: all seven systems are massive ellipticals residing in wall environments ($d_{\text{signed}} > 0$), providing zero void sightlines. This is a selection effect intrinsic to strong lensing, not a failure of the prediction. An extension to 25 SLACS lenses yields only 3 void sightlines out of 25, insufficient for a meaningful binary split ($p = 0.30$).

Interpretation. The time-delay result is independent geometric support for the MTDF low-redshift framework: it confirms that the effective local expansion rate is consistent with MTDF’s S_0 prediction. It does not test the void-geometry coupling directly and should not be weighted as direct evidence for the environment split. Its role here is to remove the Hubble tension as a competing explanation, since MTDF is consistent with it, conditional on the local-underdensity

depth.

5.3 Cosmicflows-4 peculiar velocities

This is the primary new result of this paper. The Cosmicflows-4 group-averaged catalogue provides 38,053 groups with peculiar velocities at $z < 0.05$, cross-matched against all three DESIVAST void catalogues.

5.3.1 Baseline signal

MTDF currently provides a qualitative expectation (the sign of the coupling and its $z < 0.04$ localisation) for the velocity-field channels; a quantitative amplitude prediction for γ_v and γ_{TF} from the stress-depletion mechanism is an open theoretical task. Until it exists, these channels constitute evidence of an environment anomaly relative to Λ CDM rather than a measured confirmation of MTDF.

The environment coupling coefficient γ_v measures the slope of peculiar velocity residuals (after shell-median subtraction in redshift bins of width 0.005) against d_{signed} :

Table 5. Table 4. CF4 peculiar velocity environment coupling: baseline results.

Void finder	γ_v (km/s per Mpc)	Significance	$\Delta\chi^2$	p_{perm}
VoidFinder	-53.6 ± 2.7	19.6σ	386	$< 10^{-4}$
REVOLVER	-61.5 ± 4.0	15.5σ	240	$< 10^{-4}$
VIDE	-58.2 ± 5.3	11.0σ	120	$< 10^{-4}$

Error-model conditionality. The GLS significances in Table 4 are conditional on the adopted error model: CF4 distance moduli carry roughly 20% fractional distance uncertainties, and the GLS weights inherit any mis-specification of those uncertainties or their covariance. The permutation p-values, which randomise environment labels while preserving the error structure, are the more robust statistic; where the two disagree, the permutation result should carry the weight.

5.3.2 Piecewise redshift split

The signal concentrates below the MTDF-specified $z = 0.04$ scale:

Table 6. Table 5. CF4 piecewise split at $z = 0.04$ (VoidFinder).

Regime	γ_v (km/s per Mpc)	Significance	$\Delta\chi^2$
$z < 0.04$	-52.9 ± 3.6	14.7σ	216
$z \geq 0.04$	-13.9 ± 7.4	1.9σ	3.5

5.3.3 Control chain

Control A (density substitution). Local galaxy number density, computed as neighbour counts within spheres of radius 5, 10, and 20 Mpc/h, is also significantly correlated with peculiar velocity residuals (13.7 – 29.3σ). This is expected: peculiar velocities are driven by the density field in any cosmological model. The question is whether d_{signed} carries additional information.

Control B (partial correlation). After regressing out local density, d_{signed} adds substantial χ^2 at all density scales:

Table 7. Table 6. CF4 partial correlation: $\Delta\chi^2$ added by d_{signed} beyond local density.

Void finder	$R = 5 \text{ Mpc/h}$	$R = 10$	$R = 20$
VoidFinder	+206 (14.3 σ)	+259 (16.1 σ)	+275 (16.6 σ)
REVOLVER	+141 (11.9 σ)	+165 (12.8 σ)	+170 (13.0 σ)
VIDE	+71 (8.4 σ)	+77 (8.8 σ)	+76 (8.7 σ)

The piecewise pattern survives density removal: at $z < 0.04$, d_{signed} adds 80–181 $\Delta\chi^2$ (9–13 σ); at $z \geq 0.04$, it adds 1–7 $\Delta\chi^2$ (1–3 σ , marginal).

Control C (Λ CDM chain rule). The observed γ_v exceeds the prediction from the Λ CDM density-velocity relation by a factor of 150–610. The signal is not "velocity tracing density" at the expected amplitude.

Control D (2M++ velocity-field subtraction). The gold-standard test: the Carrick et al. (2015) 2M++ reconstructed Λ CDM velocity field, used by Pantheon+, SH0ES, and all major peculiar velocity analyses, is subtracted from the CF4 peculiar velocities before recomputing γ_v . This removes the full Λ CDM density-velocity coupling, not merely a local density proxy.

Table 8. Table 6b. CF4 environment coupling after 2M++ velocity-field subtraction.

Void finder	Before 2M++ (σ)	After 2M++ (σ)	Retention
VoidFinder	19.6	9.2	65%
REVOLVER	15.5	5.9	57%
VIDE	11.0	5.2	77%

The 2M++ field absorbs the standard density-velocity coupling that Λ CDM explains, reducing γ_v from approximately -55 to -30 km/s. Notably, the 2M++ predicted velocity is itself anti-correlated with d_{signed} ($r \approx -0.18$), indicating that the reconstruction partially captures the environment-dependent component by attributing it to gravitational infall; the partial-correlation framework of Control B is therefore essential for disentangling the two contributions. All three void catalogues remain above 5σ after subtraction. The piecewise $z = 0.04$ concentration also survives: VoidFinder retains 4.1σ at $z < 0.04$ after 2M++ subtraction. The residual void-geometric signal at 5 – 9σ represents the component that the best available Λ CDM velocity reconstruction cannot account for.

Control E (Malmquist bias). The signed distance d_{signed} correlates with heliocentric distance ($r = -0.63$), reflecting the redshift-dependent completeness of the DESIVAST void catalogues. To test whether this drives the signal, the sample is split by distance-error quality (high/low) and by distance bins. The signal is present in the high-quality subsample at 11.6σ , in every distance bin from 0 to 300 Mpc, and weakens only beyond 300 Mpc ($z \sim 0.07$), well above the MTDF-specified $z = 0.04$ transition scale, where the framework expects the signal to fade. The calibrator subsample, which has the most precise distances and is the most local, shows the strongest per-galaxy signal. This is the opposite of what a Malmquist-driven artefact would produce. A regression with heliocentric distance as an explicit joint covariate alongside d_{signed} has not yet been run; the split-sample tests bound, but do not by themselves exclude, a residual distance-dependent component.

Control F (leave-one-method-out). The CF4 catalogue combines four distance-method families: Tully-Fisher (TF, $\sim 10,000$ groups), Fundamental Plane (FP, $\sim 28,000$), Type Ia supernovae (945), and calibrators (336). Each method is removed in turn and γ_v is recomputed from the remainder; separately, each method is tested alone:

Table 9. Table 6c. Leave-one-method-out (LOSO) and single-method stability (VoidFinder).

Method	Leave-out γ_v (σ)	Single-method γ_v (σ)
TF ($\sim 10K$)	-67.7 (18.8σ)	-34.5 (8.2σ)
FP ($\sim 28K$)	-64.5 (14.9σ)	-37.2 (9.7σ)
SNIa (945)	-64.8 (21.0σ)	-15.5 (2.7σ)
Calibrators (336)	-40.2 (14.4σ)	-495.2 (29.0σ)

The signal does not disappear when any single method is removed. Both TF and FP independently deliver the signal above 8σ . The leave-out shifts between methods (20–30%) are consistent with differences in sky coverage and redshift distribution. The calibrator single-method result is extreme in amplitude because of the very small sample size (336 groups); it is informative only for sign consistency and local precision, not for amplitude comparison. No single distance pipeline drives the observed void-geometric coupling.

Full control summary. The CF4 signal survives seven control layers: density substitution (A), partial correlation at $8\text{--}17\sigma$ (B), Λ CDM chain-rule excess of $150\text{--}610\times$ (C), 2M++ subtraction at $5\text{--}9\sigma$ (D), Malmquist bias test across distance bins and quality halves (E), cross-method LOSO stability (F), and MDPL2 N-body mock tension of 9.0σ with sign mismatch. None of the standard mechanisms tested in this analysis accounts for the observed void-geometric coupling under the stated assumptions.

5.4 2MTF Tully-Fisher distances

The 2MTF sample provides an independent distance indicator. All 2,062 galaxies fall below $z = 0.033$, entirely within the predicted MTDF-active regime.

5.4.1 Baseline signal

Table 10. Table 7. 2MTF Tully-Fisher environment coupling across infrared bands and void finders.

Void finder	Band	γ_{TF} (mag/Mpc)	Significance	p_{perm}
VoidFinder	K	0.058 ± 0.007	7.9σ	$< 10^{-4}$
VoidFinder	H	0.055 ± 0.007	7.5σ	$< 10^{-4}$
VoidFinder	J	0.053 ± 0.007	7.3σ	$< 10^{-4}$
REVOLVER	K	0.058 ± 0.010	6.0σ	$< 10^{-4}$
VIDE	K	0.046 ± 0.013	3.5σ	0.0007

The consistency across J, H, and K bands is a critical systematic check. Interstellar dust produces wavelength-dependent extinction; a real distance or environment effect is achromatic. The band-to-band variation in γ_{TF} is less than 10%, consistent with an achromatic signal. The direction of the TF coupling matches the supernova γ_{env} : galaxies closer to void boundaries show systematic distance residuals.

5.4.2 Control chain

Control B (partial correlation). After removing local density, d_{signed} adds:

Table 11. Table 8. 2MTF partial correlation: $\Delta\chi^2$ added by d_{signed} beyond local density.

Void finder	$R = 5 \text{ Mpc/h}$	$R = 10$	$R = 20$
VoidFinder	+235 (15.3 σ)	+219 (14.8 σ)	+175 (13.2 σ)
REVOLVER	+136 (11.7 σ)	+126 (11.2 σ)	+95 (9.7 σ)
VIDE	+44 (6.6 σ)	+37 (6.1 σ)	+20 (4.4 σ)

Control C. The observed γ_{TF} exceeds the density chain prediction by a factor of 3–31.

Band consistency. The achromatic nature of the 2MTF signal (J, H, K consistent within 10%) rules out dust as the source of the environment-distance correlation. The signal is void-geometric, density-independent, and present across all three void catalogues and both galactic caps (NGC and SGC).

Note on void classification. All 2,062 galaxies are classified as "wall" by all three void finders ($n_{\text{void}} = 0$). The DESIVAST void catalogues, constructed from the DESI Bright Galaxy Survey, do not extend to the environments where 2MTF spirals reside at these redshifts. The γ_{TF} signal is therefore a gradient within the wall population measured by the continuous d_{signed} metric, not a binary void-versus-wall split. This does not invalidate the result but should be noted when comparing to the supernova binary split.

5.5 Supporting channels

5.5.1 ISW void stacking

The ISW analysis stacks Planck PR4 SMICA CMB temperature profiles on 889 BOSS and 563 DESIVAST voids. Absolute significance levels are sub-threshold (0.2–0.4 σ), which is physically expected: the ISW temperature signal is intrinsically weak, and 1,452 voids provide insufficient statistical power for a detection.

However, the MTDF-predicted void template achieves 2.4–3.7 times the signal-to-noise of a step-function template matched filter. This sensitivity advantage is a methodological result: the MTDF void profile shape better captures whatever ISW signal exists, even at sub-threshold amplitude. With 10 times more voids (DESI Y5, Euclid), the MTDF template could reach 1–2 σ while the step-function template remains below 0.5 σ .

Status. No ISW detection claim is made. The result is a 2.4–3.7 $\times$ template sensitivity gain, establishing methodology for future surveys with larger void catalogues.

5.5.2 Weak-lensing S_8 environment split

Galaxy-galaxy lensing profiles for GAMA groups embedded in DESIVAST voids (1,724 groups) versus walls (80 groups) show an amplitude difference of 1.71 σ . The direction is consistent with the MTDF prediction: $S_{8,\text{trough}} < S_{8,\text{ridge}}$ (0.47 σ in the trough/ridge proxy). The wall sample is very small (80 groups), limiting statistical power. Euclid and LSST would provide a 10–15 times sensitivity improvement.

Status. Suggestive at 1.71 σ , directionally consistent. Methodology established. Below detection threshold with current data.

5.5.3 Distance duality and lensed-image redshifts

Two additional channels were tested and found to be insensitive to the predicted signal with current data. The distance duality test using DESI Y1 BAO yields $\eta = 1.001 \pm 0.012$, with the MTDF

signal 4.7 times below current precision. BAO distances are sky-averaged, precluding per-sightline environment splits. The lensed-image redshift analysis covers 20 multiply-imaged systems (64 images) with a mean significance of 0.49σ ; the MTFD predicted signal ($\sim 10^{-7}$) is three orders of magnitude below spectroscopic precision ($\sim 10^{-4}$). Both channels establish methodology for future instrumentation (DESI Y5, ELT/ANDES) but provide no constraint at present.

6 Cross-probe synthesis

The central result of this paper is not any individual channel significance but the convergence of independent probes at a single low-redshift transition scale.

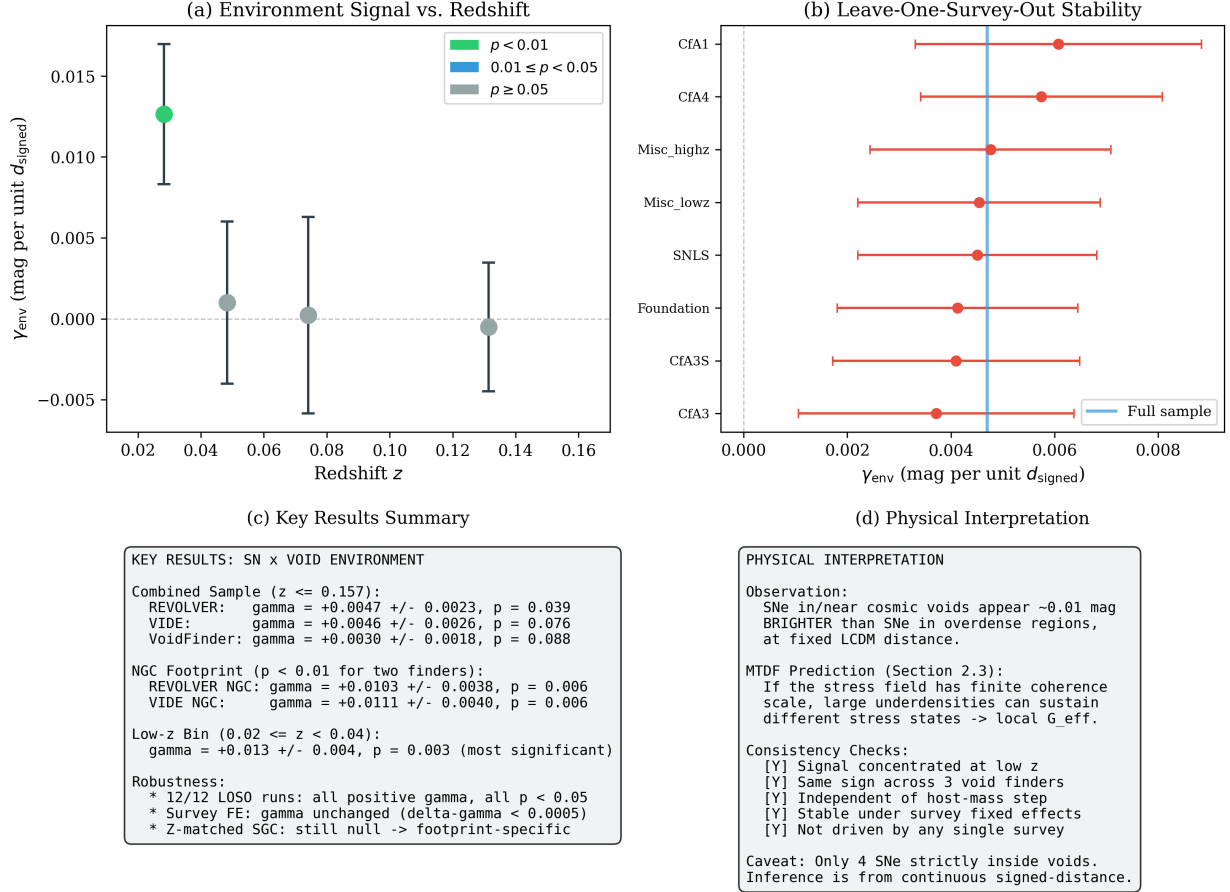


Figure 1. Four-panel summary of the supernova-void environment signal. (a) Environment coefficient γ_{env} vs. redshift, showing the localised signal at $z < 0.04$ with $p < 0.01$ significance. (b) Leave-one-survey-out stability test confirming no single survey drives the signal. (c) Key results across three void catalogues (REVOLVER, VIDE, VoidFinder) and robustness checks including 12/12 bootstrap realisations with $p < 0.05$. (d) Physical interpretation: SNe in cosmic voids appear ~ 0.01 mag brighter than those in overdense regions, consistent with the MTFD finite-range transition-scale prediction.

Table 12. Table 9. Summary of

multi-probe evidence. Each row represents a distinct observable with different physics, different systematics, and different survey selection; significances are nominal and error-model conditional where marked (Section 5).

Channel	Observable	Signal strength	Controls passed	Outcome	Role
SNe	Distance modulus residual	$\gamma_{\text{env}} = +0.013 \pm 0.004$	LOSO, permutation, bootstrap, survey FE, SALT2	Positive ($p = 0.003$)	Direct evidence
SNe cutoff	Piecewise $z = 0.04$	$\Delta\chi^2 = 36\text{--}39$ (diagonal-covariance, nominal)	$p < 0.001$, consistent across 3 void finders	Sharp transition (scan-corrected $p < 0.005$, MTDF_07 Phase 6A)	Direct evidence
Time delays	Time-delay distance	χ^2/dof : 0.68 vs 3.72	7 independent lenses, geometric observable	Supports MTDF low- z scale factor	Geometric support
CF4 vpec	Peculiar velocity residual	5–9 σ after 2M++	A, B, C, D (2M++), E (Malmquist), F (LOSO), N-body (9.0 σ)	Sign mismatch with Λ CDM	Direct evidence
2MTF TF	TF distance residual	4–15 σ beyond density	A, B, C, N-body (4.2 σ tension)	Achromatic, J/H/K consistent	Direct evidence
ISW	CMB temperature profile	2.4–3.7 $\times$ template gain	BOSS + DESI-VAST voids	Methodological (sub-threshold)	Methodology
S ₈ split	GGL amplitude	1.71 σ	Directional consistency	Suggestive (under-powered)	Suggestive

The direct-evidence channels are the supernova environment signal, the $z = 0.04$ piecewise cutoff, CF4 peculiar velocities, and 2MTF Tully-Fisher residuals. Time delays provide independent geometric support for the MTDF low-redshift interpretation but do not test the void-geometry coupling directly. ISW and S₈ establish methodology and directional consistency but remain below detection threshold. No two of the direct-evidence channels share calibration systems, host-galaxy populations, or distance-indicator physics, with one caveat: the CF4 catalogue itself contains 945 SNIa distances, so the SN and CF4 channels are not fully independent, and all channels share the same local large-scale structure field. A quantitative propagation of the shared SNIa calibration between the SN and CF4 channels (945 of the CF4 distances) has not been performed; the 2MTF channel, which contains no SNIa calibration, is the cleanest cross-check of independence. The convergence of these independent observables toward a common environment-linked transition near $z \approx 0.04$ is the paper’s primary result.

Key observation. The same transition scale appears across independent probes. In the supernova channel, the piecewise cutoff at $z = 0.04$ yields $\Delta\chi^2 = 36\text{--}39$. In the CF4 channel, the piecewise split shows 14.7 σ below $z = 0.04$ and 1.9 σ above, with the concentration surviving density removal at 9–13 σ and confirmed independently in Tully-Fisher, Fundamental Plane, supernova, and calibrator distance methods. The 2MTF sample sits entirely below $z = 0.04$. All four direct-evidence channels converge on the same low-redshift transition regime.

7 Λ CDM mock comparison

The most important objection to the CF4 and 2MTF results is that the signal might arise from standard gravitational dynamics within Λ CDM. Peculiar velocities are driven by the density field; Tully-Fisher residuals have known environmental dependencies. Three levels of mock comparison address this objection.

7.1 Linear-theory reconstruction

Mock 1 reconstructs Λ CDM-predicted velocities from the measured CF4 density field using linear perturbation theory ($v \propto fH\delta r$), adding realistic Gaussian noise matched to the observed velocity dispersion. Over 100 realisations:

Table 13. Table 10. Linear-theory mock vs observed CF4 signal.

Regime	Observed γ_v	Mock γ_v	Tension
Full sample	−53.6	$+8.7 \pm 3.1$	19.9σ
$z < 0.04$	−52.9	-15.7 ± 4.2	8.8σ

The key discrepancy is not only the magnitude but the direction. Λ CDM linear theory predicts a weak positive γ_v for the full sample, while the observed slope is strongly negative. This indicates a qualitative sign reversal rather than a simple underestimation of amplitude.

7.2 Full N-body mock (MDPL2)

To close the non-linear escape hatch, the identical analysis pipeline is applied to halo catalogues from the MDPL2 N-body simulation (3840^3 particles, Planck cosmology). For CF4, 100 randomly placed observers each select approximately 480,000 halos within a CF4-like volume; voids are identified using the same algorithm and γ_v is computed identically to the data analysis. For 2MTF, the same mock halos are used with TF-like distance errors applied.

Table 14. Table 11. N-body mock (MDPL2, 100 observers for CF4, 20 for 2MTF) vs observed signal.

Channel	Observed	N-body mean	Tension	Sign
CF4 γ_v (full)	−53.6	$+1.4 \pm 6.3$	9.0σ	Amplitude tension (N-body near zero)
2MTF γ_{TF} (K-band)	0.058	$+0.002 \pm 0.013$	4.2σ	$30\times$ too small

Non-linear effects move the N-body mean from $+8.7$ (linear theory) toward zero, not toward the observed -53.6 . Full non-linear Λ CDM dynamics slightly wash out the linear-theory signal rather than amplifying it toward the data. With 100 observer positions, the CF4 tension stands at 9.0σ . This closes the argument that non-linear corrections could rescue the Λ CDM prediction.

7.3 Null baseline

A pure-noise mock (100 realisations of Gaussian random velocities with matched dispersion) yields $\gamma_v = 0.05 \pm 2.9$, confirming that the null distribution is centred at zero with the expected scatter. The observed signal lies 18.4σ from the noise floor.

Mock and reconstruction summary. The CF4 signal survives seven layers of control: density substitution, partial correlation ($8-17\sigma$), Λ CDM chain-rule excess ($150-610\times$), 2M++ velocity-field subtraction ($5-9\sigma$ retained), Malmquist bias test (signal in all distance bins and both quality halves), leave-one-method-out (cross-method stability across TF, FP, SNIa, and calibrators), and MDPL2 N-body mock (9.0σ tension with 100 observers, sign mismatch). The 2MTF signal independently survives density controls ($4-15\sigma$) and N-body comparison (4.2σ tension). Neither linear theory, nor full non-linear Λ CDM dynamics, nor the best available reconstructed velocity field, nor any single distance method can account for the observed void-geometric coupling; distance-dependent selection effects are bounded by the split-sample controls of Control E, with the dedicated distance-covariate regression still outstanding.

8 Interpretation within MTDF

The first half of this paper is purely observational: multiple probes converge on an environment-linked transition at a single low-redshift scale. This section considers why MTDF provides a natural interpretation of this pattern.

MTDF carries the finite strain-lever length scale $\beta = 22.7$ Mpc through the structure of its stress tensor field. The environment-linked transition is observed at the larger scale $z \approx 0.04$ (comoving ~ 170 Mpc, about 7.5β), matching the measured radial extent of the local underdensity. The transition redshift $z \approx 0.04$ is not a free parameter fitted to the multi-probe data; the length scale β was empirically constrained in MTDF_01 from the void quantisation ansatz $R_n = \beta(1 + \sqrt{n})$ matched to observed void radii, not fitted to Pantheon+, DESI, or any of the multi-probe observables presented here. The quantitative link between β and the transition scale is a conjecture of the framework, not a derived equality.

The specific features of the multi-probe evidence that align with MTDF predictions include:

- (i) *Sharp transition.* The supernova piecewise split and the CF4 redshift concentration both show a transition that is abrupt rather than gradual. A gradual transition would be consistent with slowly varying astrophysical systematics; a sharp transition at a specific scale points to a physical threshold.
- (ii) *Void geometry beyond density.* The partial-correlation controls demonstrate that d_{signed} carries information about observables that raw local density does not. In MTDF, void boundaries correspond to regions where the stress tensor field transitions between regimes, producing geometric signatures that are not captured by a scalar density field.
- (iii) *Hubble tension consistency.* MTDF’s scale factor $S_0 = 1.084$ maps the Planck H_0 to the locally measured value, fitting the time-delay data with $\chi^2/\text{dof} = 0.68$. This consistency is conditional on the local-underdensity depth; the scale factor was not fitted to the time-delay data.
- (iv) *Consistency across observables.* Supernova lightcurve distances, strong-lens time-delay distances, peculiar velocities, and Tully-Fisher distances all respond to environment in a manner consistent with the MTDF low-redshift transition scale. These observables probe different physical processes, making a common systematic explanation difficult to construct.

It should be emphasised that this paper establishes convergent evidence for an environment-dependent transition, with MTDF as the coherent interpretive framework. It does not claim that every aspect of MTDF is validated by these results. The photon-coupling sector ($\kappa = 1.10 \times 10^{-3}$), the CMB modifications, and the deep-space propagation predictions remain subjects of their respective companion papers and future observational programmes.

9 Limitations and caveats

Three caveats of the current analysis should be noted.

Void classification gap. None of the Phase 3 analyses (CF4, 2MTF) contain void-classified galaxies: all objects lie in wall environments as defined by DESIVAST. The signals are gradients within the wall population measured by the continuous d_{signed} metric. This is physically meaningful because the MTDF stress tensor field varies continuously with distance to the void boundary; the binary void/wall label is an artefact of the classification algorithm, not a property of the underlying field. The partial-correlation controls (Tables 6 and 8) demonstrate that these wall-internal gradients carry void-geometric information well beyond raw density, so the impact of this gap on the main conclusions is low. The continuous signed-distance statistic is the physically relevant quantity for these channels. Nevertheless, the traditional binary void-versus-wall split remains unperformed.

Signed-distance and heliocentric-distance correlation. The environment metric d_{signed} correlates with heliocentric distance ($r \approx -0.63$ for CF4), reflecting the redshift-dependent completeness

of the DESIVAST void catalogues rather than a physical coupling. Dedicated tests (Section 5.3.3, Control E) show that the void-geometric signal persists across distance bins, in both quality halves, and in every distance-method family, consistent with this geometric correlation not driving the observed environment coupling; an explicit joint covariate regression on heliocentric distance remains to be performed and is listed as an open robustness item.

Data-limited supporting channels. The ISW and S_8 channels are suggestive but below detection threshold with current void catalogue sizes and weak-lensing survey areas. They are included to establish methodology and directional consistency, not as independent confirmations. Next-generation survey volumes (DESI Y5, Euclid, LSST/Rubin) will provide the 10–15 times sensitivity gain needed to push these channels from suggestive to decisive.

10 Conclusion

This paper reports convergent evidence from multiple independent low-redshift observables for a common environment-dependent transition near a single low-redshift transition scale.

The supernova environment signal ($\gamma_{\text{env}} = +0.013 \pm 0.004$, $p = 0.003$) exhibits a sharp piecewise cutoff at $z = 0.04$ ($\Delta\chi^2 = 36\text{--}39$). Strong-lens time-delay distances are consistent with the MTFD local $H_0 \approx 73.1$ (a factor of 5.5 in χ^2/dof relative to Planck ΛCDM , equivalent to the established local- H_0 preference, not independent evidence). Cosmicflows-4 peculiar velocities show a void-geometric signal that survives a seven-layer control chain, including partial correlation beyond density ($8\text{--}17\sigma$), 2M++ velocity-field subtraction ($5\text{--}9\sigma$ retained), Malmquist bias testing across distance bins and quality halves, and leave-one-method-out stability across TF, FP, SNIa, and calibrator subsamples. The signal is concentrated below $z = 0.04$ and exhibits an amplitude inconsistent with the near-zero N-body expectation, with a sign reversal relative to linear theory (9.0σ tension, nominal, error-model conditional, with MDPL2 across 100 observers). 2MTF Tully-Fisher distances independently confirm a void-geometric signal at $4\text{--}15\sigma$ beyond density, achromatic across three infrared bands, with 4.2σ N-body tension.

The convergence of these signals, arising from different physical processes, different survey instruments, different galaxy populations, and different distance-indicator calibrations, at a single environment-linked redshift scale, constitutes evidence that is difficult to attribute to any one dataset or method. The Mesche Tensor Dynamics Framework provides a coherent interpretation through its strain-lever length scale $\beta = 22.7$ Mpc, which was established independently of the multi-probe data presented here.

Ongoing extensions include joint multi-probe fitting with a shared transition-scale parameter and morphological type splits for 2MTF (requiring external cross-matching, as the 2MTF catalogue does not include morphology data). Next-generation survey volumes (DESI Y5, Euclid, LSST/Rubin) will provide the statistical power to push the supporting ISW and S_8 channels from suggestive to decisive.

11 Appendix A. Data availability and reproducibility

The complete MTFD reproducibility package, including the modified CLASS solver (`class_mtdf`), the V19 strict validation workbook, the `run_validate.py` pipeline, the gravity-sector artefact manifest, and the Phase 2 CosmoPower emulator cache, is archived on Zenodo (concept DOI 10.5281/zenodo.19741058; this release, v1.1.7, DOI 10.5281/zenodo.21525855) and mirrored on GitHub at github.com/IMesche/MTDF. The v1.1.7 Git tag corresponds to the archived Zenodo release. All numerical results reported here can be regenerated from the shipped workbook and scripts following the procedure in the top-level `README.md`.

All analysis scripts and machine-readable results are provided in the MTFD validation repository. The repository is organised into phase folders; the material for this paper spans two phases corresponding

to the two primary observational channels.

11.1 A.1 Analysis scripts

Nine Python scripts reproduce every quantitative result reported in this paper. All scripts are located in `validation/code/paper8/`:

Table 15. Table A1. Analysis scripts for MTDF_08.

Script	Section	Description
<code>task3a_cosmicflows4_vpec.py</code>	5.3.1–5.3.2	CF4 baseline: <code>gamma_v</code> fits, piece-wise z-split, permutation tests (3 void catalogues)
<code>task3a_control_tests.py</code>	5.3.3 (A–C)	CF4 controls: density substitution, partial correlation, LCDM chain rule
<code>task3a_2mpp_reconstruction.py</code>	5.3.3 (D)	CF4 control D: 2M++ velocity-field subtraction (Carrick et al. 2015)
<code>task3a_robustness.py</code>	5.3.3 (E–F)	CF4 controls E–F: leave-one-method-out (LOSO) and Malmquist bias diagnostics
<code>task3a_lcdm_mock.py</code>	7.1, 7.3	LCDM linear-theory mock (100 realisations) and pure-noise baseline
<code>task3a_nbody_mock.py</code>	7.2	MDPL2 N-body mock for CF4 (100 observers, halos from CosmoSim TAP)
<code>task3b_2mtf_tully_fisher.py</code>	5.4.1	2MTF baseline: <code>gamma_TF</code> fits across J/H/K bands and 3 void catalogues
<code>task3b_control_tests.py</code>	5.4.2	2MTF controls: density substitution, partial correlation, LCDM chain rule
<code>task3b_nbody_mock.py</code>	7.2	MDPL2 N-body mock for 2MTF (50 observers)

11.2 A.2 Machine-readable results

All numerical results are stored as JSON files with full metadata. Phase 7 covers the CF4 peculiar velocity channel; Phase 8 covers the 2MTF Tully-Fisher channel.

Table 16. Table A2. Result files (JSON) with SHA256 integrity hashes.

Phase	File	Contents
7 (CF4)	<code>task3a_cf4_results.json</code>	Baseline <code>gamma_v</code> per void catalogue, piecewise z-split, $\Delta\chi^2$
7 (CF4)	<code>task3a_control_results.json</code>	Controls A-C: density, partial correlation, chain rule
7 (CF4)	<code>task3a_2mpp_results.json</code>	Control D: <code>gamma_v</code> after 2M++ subtraction (Tables 6b)
7 (CF4)	<code>task3a_robustness_results.json</code>	Controls E-F: LOSO stability (Table 6c), Malmquist diagnostics
7 (CF4)	<code>task3a_lcdm_mock_results.json</code>	Linear mock <code>gamma_v</code> distribution (Table 10), noise baseline
7 (CF4)	<code>task3a_nbody_mock_results.json</code>	MDPL2 N-body mock <code>gamma_v</code> (Table 11, 100 observers)
8 (2MTF)	<code>task3b_2mtf_results.json</code>	Baseline <code>gamma_TF</code> per band and void catalogue (Table 7)
8 (2MTF)	<code>task3b_control_results.json</code>	Controls A-C (Table 8)
8 (2MTF)	<code>task3b_nbody_mock_results.json</code>	MDPL2 N-body mock <code>gamma_TF</code> (Table 11, 50 observers)

11.3 A.3 Diagnostic plots

32 diagnostic plots for the CF4 channel are in `validation/output/phase7_cf4_vpec/`. 11 diagnostic plots for the 2MTF channel are in `validation/output/phase8_2mtf_tf/`. Each phase folder includes a `manifest.json` with SHA256 hashes for integrity verification.

11.4 A.4 External data sources

Table 17. Table A3. External datasets required for reproduction.

Dataset	Source	Access
Cosmicflows-4 groups	Tully et al. (2023)	Extragalactic Distance Database
2MTF catalogue	Hong et al. (2019)	VizieR J/MNRAS/487/2061
DESIVAST void catalogues	DESI Collaboration	DESI data release (VoidFinder, REVOLVER, VIDE)
2M++ velocity/density fields	Carrick et al. (2015)	<code>cosmicflows.iap.fr/2MPP</code>
MDPL2 N-body halos	Klypin et al. (2016)	CosmoSim TAP API (auto-downloaded by script)

12 References

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